

CHE 381/382
Chemical Engineering Laboratory I & II Syllabus

Instructor:

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Course Texts:

Unit Operations of Chemical Engineering
W.L. McCabe, J.C. Smith, P. Harriott
7th Edition, McGraw-Hill, 2005.
Optional: Writing Styles and Standards in
Undergraduate Reports, J. Donnell
S. Jeter, C. MacDougall, J. Snedeker,
3rd Ed. College Publishing, 2016.

Introduction:

The laboratory component of an undergraduate education is critical to the professional success of scientists and engineers. CHE381 and CHE382, also known as Unit Operations Laboratory I and II, provides students with an exceptional hands-on opportunity to work with physical equipment in a laboratory environment. Students, organized into small groups, will conduct experiments, write a pre-lab plan and final lab report, and present their work to their colleagues.

Student Learning Objectives

Provide students with a hands-on opportunity to work with chemical engineering process equipment in a laboratory environment. Students will learn to solve complex, open-ended engineering problems in a team setting, integrate knowledge and skills from prior chemical engineering courses, design and conduct realistic experiments, analyze data, draw concrete conclusions and make relevant recommendations, communicate findings effectively through the written and oral word, and apply safety practices in a laboratory environment.

List of Experiments

Each team will perform the following experiments in CHE381 and CHE382:

- | CHE381 | CHE382 |
|------------------------------------|-----------------------------|
| 1. Viscosity: Non-newtonian fluids | 1. Evaporation |
| 2. Distillation | 2. Ultrafiltration |
| 3. Fluid Flow in Pipe Systems | 3. Heat Transfer |
| 4. Fluidized Bed | 4. Gas Absorption |
| 5. Process Control | 5. Liquid-Liquid Extraction |
| 6. Membrane Separation | 6. Humidification |

The Blackboard website contains the most up-to-date manuals for carrying out the various unit operations experiments (pdf files). Useful information on each lab is provided in Blackboard to help aid in successfully preparing and carrying out the experiments, and to write the final report.

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Safety in the Laboratory

First and foremost, safety is an integral component in all laboratory environments, and certainly of all experiments and industrial processes. The Chemical Engineering Laboratory Safety document is your guidance in this regard. You must a) read, sign and submit this document stating in effect that you have read, understood, and will comply with the regulations contained therein. The safety document must be completed before you begin any lab work in this course. It is a requirement that students in the lab take a proactive approach to safety and remind others of safety practices when necessary. The laboratory is an ideal place to develop safety awareness and skills; those who embrace the safety culture in the workplace will be highly regarded and rewarded professionally and financially in their future careers.

SAFETY GLASSES: Safety glasses must always be worn in the lab. Exception: Safety glasses are not required at the Study Tables in the back of the lab.

Pick up a pair when you enter the laboratory in the morning and put them on.

Compliance will be monitored by the instructor and TA during the lab period.

Non-compliance will be recorded.

3 instances of non-compliance will result in a zero on the Individual Assessment grade.

Laboratory Access and Attendance

Student access to the Unit Operations Laboratory (150 EIB) will coincide with the scheduled class period; every Tuesday from 9:00am to 12:30 pm (Group A) and 1:00pm – 4:30 pm (Group B) throughout the semester. The nature of laboratory studies is such that they are time and activity intensive; *please be on-time so that the laboratory can begin promptly.*

The experiments are designed to be performed in two class periods (pre-lab and lab). If additional time is needed, permission from the instructor is required to perform additional lab work.

Attendance is mandatory for the full lab period. Attendance will be recorded for each lab period and will be one of the criteria used in the individual assessment grade. Arriving 15 minutes or later will result in a late mark (L). Arriving after 50 minutes or later will result in an absent mark (A) for that lab period. Please notify the TA or instructor before class if you cannot make the lab period. Excused absences will be considered on an individual basis. Two (2) or more unexcused absences will result in a zero on the Individual Performance grade.

Teamwork

Students will be organized into lab teams. Each team will conduct experiments, write pre-lab and final reports, and present their work to their colleagues. All team members are expected to participate jointly in planning and conducting the experiments, evaluating experimental data, analyzing results, writing reports and in the oral presentations. The team should also decide on a team leader. Team member contributions will be tracked by each student filling out the Team Member Contribution form after each lab experiment (form is in Blackboard). The forms will be attached to the final lab report and will be used as part of the Individual Performance grade.

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Lab and Assignment Cycle

Two lab periods are allocated for each experiment as follows:

Period 1 (Lab Prep): This lab period is devoted to becoming familiar with the equipment and preparing an experimental plan that will be carried out in Week 2. Tasks for individual team members will be assigned. The team will review and understand the basic theory of the experiment, formulate a detailed step-by-step experimental procedure, conduct a safety analysis on the equipment, *carry out preliminary measurements*, and write and submit a Pre-Lab Plan at the end of the lab period. Format guidelines for writing the Pre-Lab Plan are available in Blackboard.

Period 2: This lab period is devoted to performing the experiment; collecting the data and analyzing the results, formulating conclusions and recommendations, and investigating errors in measured variables. A **Final Lab Report** will be written.

The Final Report will be due the following Monday at 5:00 PM. Format guidelines for writing the Final Lab Report are available in Blackboard. Several types of final report styles may be used for different experiments (short narrative, long narrative). In addition, preparation for the team oral presentation will be initiated.

Next Period 1: Before starting the next experiment, the team will make a 10 minute presentation to the group that will perform the same experiment the following week. The presentation should be brief but concise; it should describe motivation, theory, experimental procedures, experimental errors, results and conclusions, and lessons learned (possible improvements). The presentation will be given in front of the apparatus, and the presentation given to the TA's afterwards for grading. A template for the oral presentation is in Blackboard.

Safety Assignments I and II: A safety assignment and quiz will be conducted at the beginning of the semester to provide additional safety learning and comprehension. Students will also be required to complete one safety tutorial from the SACHE (Safety and Chemical Engineering Education) website (www.sache.org). Further details will follow during the semester.

Grade:

20% Lab Prep Report; 40% Final Lab Report; 10% Oral Presentation;
10% Safety Quiz; 20% Individual Performance Assessment

Assessment of written reports will be based on the following attributes:

- Completeness (inclusion of all relevant content and report sections specified)
- Organization (orderly and logical flow of material), clarity, conciseness including report length, and grammatical correctness.
- Technical correctness of calculations
- Technical correctness of analysis, results, and conclusions
- Initiative (original work to more efficiently achieve objectives or go beyond basic objectives).

Assessment of oral reports will be based on the following attributes:

- Overall presentation quality
- Accuracy and clarity of statements
- Quality of data presentation and analysis
- Degree of technical comprehension

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A Note on Plagiarism

Plagiarism is defined as “the unacknowledged inclusion of someone else’s words, structure, ideas, or data.” To avoid even the appearance of plagiarism, one must learn when and how to cite sources correctly. One helpful website is “Plagiarism.org”. Plagiarism will not be tolerated and will result at a minimum in a zero for the report or presentation.